Concave & Convex Functions

Exam code: 9MA0

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Concave & convex functions

How do I use second derivatives to determine if a curve is concave or convex?

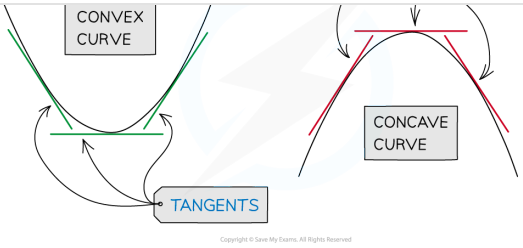
- The **second order derivative** (or simply **second derivative**) is used to help determine the nature of a stationary point
- At A level you need to be able to use the second derivative to determine if a function is **convex** or **concave** on a given interval
  - Where a function is **convex** its graph 'curves up' and its tangent lines lie below the graph
  - Where a function is **concave** its graph 'curves down' and its tangent lines lie above the graph

Help me understand

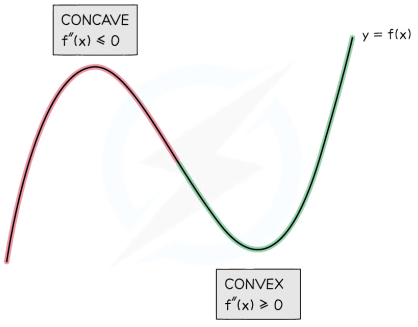
Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



- It's the sign of the second derivative (+ or –) that's important:
  - A function **f(x)** is **convex** on a given interval if and only if **f''(x) ≥ 0** for every value of ***x*** in the interval
  - A function **f(x)** is **concave** on a given interval if and only if **f''(x) ≤ 0** for every value of ***x*** in the interval



Examiner Tips and Tricks

- To get full marks in an exam your working needs to include the condition for a function to be convex or concave.

Worked Example

$f(x) = 2x^3 - 6x^2 + 13x - 7$ is convex.

$f'(x) = 6x^2 - 12x + 13$ ← FIND FIRST DERIVATIVE

$f''(x) = 12x - 12$ ← FIND SECOND DERIVATIVE

$f(x)$ IS CONVEX WHEN $f''(x) \geqslant 0$ ← STATE CONDITION FOR FUNCTION TO BE CONVEX

$12x - 12 \geqslant 0$ ← SOLVE INEQUALITY

$12x \geqslant 12$

$x \geqslant 1$

$f(x)$ IS CONVEX ON THE INTERVAL $[1, \infty)$

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